

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (Minor) (CBCS) Degree Examinations

May 2024

EC2005-1 – PROGRAMMING MICROCONTROLLERS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Minimum address lines required to manage 2 Kbyte memory chip is
 A) 11 bit B) 10 bit
 C) 12 bit D) 9 bit
- The User application program is stored in ----- memory of MCU
 A) Program ROM B) RAM
 C) EEPROM D) EPROM
- The general-purpose registers (R0-R31) of ATMEGA controllers are mapped to
 A) Program Memory (ROM) B) Data Memory (RAM)
 C) CPU D) ALU
- ATEMGA16 series is an example of
 A) 16 bit RISC B) 8 bit CISC
 C) 8 bit RISC D) 32 bit RISC
- ADD R16,R17 performs
 A) $R17 = R16 + R16$ B) $R17 = R16 + R17$
 C) $R16 = R16 + R17$ D) $R16 = R17 + R17$
- The flag reflects the result of an arithmetic or logic operation as zero or non zero is
 A) Over flow B) Zero
 C) Carry D) Negative
- Show the status of the C, H, and Z flags after the addition of 0x9C and 0x64 in the ATMEGA MCU (Initial values of C, H and Z are 0)
 A) $C = 1, H = 1$ and $Z = 1$ B) $C = 1, H = 0$ and $Z = 1$
 C) $C = 1, H = 1$ and $Z = 0$ D) $C = 0, H = 1$ and $Z = 1$
- The status of the Z flag during the execution of the following program:
 LDI R20,4
 DEC R20
 DEC R20
 DEC R20
 DEC R20
 A) $Z=1$ B) $Z=0$
 C) $Z=4$ D) $Z=2$
- The largest number that can be loaded into the GPRs is _____ in hex.
 A) 0x99 B) 0XFFF
 C) 0XFF D) 0X999
- LDS R19,0x560 instruction is an example of
 A) Direct Addressing B) Indirect Addressing
 C) Immediate Addressing D) Register Addressing

- 11.** Find the number of times the following loop is performed:
- ```
LDI R20, 200
BACK: LDI R21, 100
HERE: DEC R21
BRNE HERE
DEC R20
BRNE BACK
```
- A) 20000                      B) 300  
C) 30000                      D) 600
- 12.** The oscillator frequency if the machine cycle = 0.125 μs.
- A) 80 MHz                      B) 8 MHz  
C) 4 MHz                        D) 40 MHz
- 13.** Find the size of the delay of the code snippet below if the crystal frequency is 10 MHz:

We have the following machine cycles for each instruction of the DELAY subroutine:

|        |                  | Instruction Cycles |
|--------|------------------|--------------------|
|        | .DEF COUNT = R20 | 0                  |
| DELAY: | LDI COUNT, 0xFF  | 1                  |
| AGAIN: | NOP              | 1                  |
|        | NOP              | 1                  |
|        | DEC COUNT        | 1                  |
|        | BRNE AGAIN       | 2/1                |
|        | RET              | 4                  |

- A) 120.0  $\mu$ s  
B) 128.0  $\mu$ s.  
C) 128.0 ms.  
D) 1.28  $\mu$ s.
14. What is the result in R20 after the execution of following instruction  
LDI R20,0x35  
ANDI R20,0x0F  
A) 0xF5  
B) 0x35  
C) 0x44  
D) 0x05
15. What is the value stored in R20 after the execution of following  
LDI R20,0x04 ;  
ORI R20,0x30  
A) 0x34  
B) 0x00  
C) 0x44  
D) 0x33
16. DDRC register in ATMEGA is used to control the  
A) PORTC interrupt  
B) PORTC out value  
C) PORTC IN value  
D) PORTC I/O direction
17. LDI R20, 10  
OUT PORTB, R20  
Value in PORTB is  
A) 0x10  
B) 0x0A  
C) 0x20  
D) 0x11
18. The data type *Char* in C holds  
A) 8 bit value  
B) 16 bit value  
C) 32 bit value  
D) 10 bit value
19. DDRB = 0xF0; makes  
A) PORTB(0-3) Output, PORTB(4-7) Input  
B) PORTB output  
C) PORTB Input  
D) PORTB(0-3) Input, PORTB(4-7) Output
20. DDRB = DDRB & ~(1<<4); In C Makes  
A) bit 4 of Port B is input  
B) bit 4 of Port B is output  
C) bit 1 of Port B is input  
D) bit 1 of Port B is output

**PART - B: DESCRIPTIVE ANSWER QUESTIONS****Unit – I**

|                                                                                                                                                                                                                                                                                                                | Marks | BT* | CO* | PO* |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) Draw the internal architecture of ATMEGA series microcontroller and list its features                                                                                                                                                                                                                    | 6     | L*2 | 1   | 1   |
| b) A switch is connected to pin PC7. Using the SBRS instruction, write a program to check the status of the switch and perform the following:<br>(a) If switch = 0, send letter 'N' to Port D.<br>(b) If switch = 1, send letter 'Y' to Port D.                                                                | 10    | L3  | 2   | 1   |
| 2. a) Distinguish between microprocessor and Microcontroller system                                                                                                                                                                                                                                            | 6     | L2  | 1   | 1   |
| b) Write a assembly language program to get the x value from Port B and send $x^2 + 2x + 3$ to Port C. Assume PB3–PB0 has the x value of 0–9. Use a look-up table instead of a multiply instruction.                                                                                                           | 10    | L3  | 2   | 1   |
| 3. a) How data memory RAM is organized in ATMEGA series microcontroller ?                                                                                                                                                                                                                                      | 6     | L2  | 1   | 1   |
| b) Assume that the lower three bits of Port C are connected to three switches. Write a assembly language program to send the following ASCII characters to Port D based on the status of the switches.<br>000 --'0'<br>001 --'1'<br>010 --'2'<br>011 --'3'<br>100 --'4'<br>101 --'5'<br>110 --'6'<br>111 --'7' | 10    | L3  | 2   | 1*  |

**Unit – II**

|                                                                                                                                                                                                                                                                                      |    |    |   |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 4. a) List and explain any 3 data types used in embedded C programming for AVR microcontrollers                                                                                                                                                                                      | 6  | L2 | 3 | 1 |
| b) Write an AVR C program to toggle all the bits of Port B continuously with a 100 ms delay. Assume that the system is ATmega 328 with XTAL = 8 MHz.                                                                                                                                 | 10 | L3 | 4 | 1 |
| 5. a) Write an AVR C program to read pins 1 and 0 of Port B and issue an ASCII character to Port D according to the following table:<br>pin1 pin0<br>0 0 send '0' to Port D (notice ASCII '0' is 0x30)<br>0 1 send '1' to Port D<br>1 0 send '2' to Port D<br>1 1 send '3' to Port D | 10 | L3 | 4 | 1 |
| b) Illustrate any three bitwise operators with an example in embedded C                                                                                                                                                                                                              | 6  | L2 | 3 | 1 |
| 6. a) Write an AVR C program to convert packed BCD 0x29 to ASCII and display the bytes on PORTB and PORTC.                                                                                                                                                                           | 8  | L3 | 3 | 1 |
| b) Write an AVR C program to convert 11111101 (FD hex) to decimal and display the digits on PORTB, PORTC, and PORTD.                                                                                                                                                                 | 8  | L3 | 4 | 1 |

**Unit – III**

|                                                                                                                                                                   |    |    |   |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 7. a) An LED is connected to PC4. Assuming XTAL = 8 MHz, write a assembly or C program that toggles the LED once per second. Use timer for generating delay       | 10 | L3 | 5 | 1 |
| b) Illustrate the basic functionality and usage of timers in ATMEGA series Microcontroller                                                                        | 6  | L2 | 5 | 1 |
| 8. a) Assume that the INT0 pin is connected to a switch that is normally high. Write a assembly or C program that toggles PORTC.3 whenever the INT0 pin goes low. | 10 | L3 | 5 | 1 |
| b) What are the procedure to access external interrupts in AVR microcontroller                                                                                    | 6  | L2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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